

AirSense Pakistan

Intelligent Campus Environmental Decision Platform

Dual-Campus Pilot Deployment, Operational Decision Support and Expansion Blueprint

A Strategic Business Proposal for Budget Approval and Institutional Partnership

SUBMITTED TO

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SUBMITTED BY (COIL AI TEAM)

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Mr. Sajid (Head of CSSE / AI, Karachi Campus)

Collaborative Initiative: Developed during COIL AI Program | August 2026

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PKR 18,676 Baseline Request	PKR 20,000 Ceiling with Contingency	578,592 Hrs 10-Yr Verified Data	R2 = 0.8338 High ML Precision
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1. Executive Summary and Business Intent

AirSense Pakistan is an intelligent, multi-campus environmental decision support platform developed under the **COIL AI** collaborative research initiative between **Beaconhouse International College (BIC) Islamabad** and **BIC Karachi**. The platform transforms hyper-local particulate matter (PM2.5) and meteorological telemetry into predictive, automated operational directives for campus management.

This proposal requests formal approval for a one-time capital hardware investment of **PKR 18,676** (with a rounded project ceiling of **PKR 20,000** inclusive of a 5% contingency reserve and power management alternatives) to deploy an edge sensing station at BIC Islamabad in parallel with BIC Karachi. Sourced exclusively through authenticated Pakistan specialist vendors (Embedded Studio, Digilog.pk, Electrobes, A.E Solution, and Clopal Online), this initiative creates Pakistan's first cross-campus predictive air quality network.

Financial Guarantee: The baseline hardware subtotal is strictly verified at PKR 17,787 with an 889 PKR contingency buffer (total PKR 18,676). **Zero recurring cloud fees, subscriptions, or import duties** are incurred. BIC retains complete ownership of all hardware, datasets, and calibrated AI models.

2. Strategic Background and Collaborative Origin

Conceived during the **COIL AI** program, AirSense was engineered through structured pair collaboration between computer science and artificial intelligence students and faculty across Islamabad and Karachi. While traditional environmental monitors only report historical data, AirSense operates at the **executive decision layer**: forecasting pollution surges 1 to 24 hours in advance and translating complex data into immediate administrative actions (e.g. throttling HVAC fresh-air intake, rescheduling outdoor athletics, and broadcasting student health advisories).

BIC Islamabad (Flagship Hub)	BIC Karachi (Coastal Sister Hub)
• Project Lead: Munim Qureshi • Supervisor: Ms. Sahifa Alam (Head of CSSE/AI) • Climate Profile: Foothill basin, severe winter smog inversions, and diurnal traffic trapping. • Primary baseline training and administrative hub.	• Project Lead: Areesha Aqeel • Supervisor: Mr. Sajid (Head of CSSE/AI) • Climate Profile: Coastal marine environment, high relative humidity, and sea-breeze dispersion. • Validates cross-regional algorithm generalization.

3. 10-Year Continuous Ground-Truth & Machine Learning Engine

To ensure commercial-grade forecasting accuracy, AirSense models are not trained on toy datasets. The platform is pre-trained on **578,592 continuous hours (2015 to 2025)** combining official US Embassy/Consulate Met One BAM-1020 regulatory stations, ECMWF ERA5 hourly meteorological reanalysis, and Copernicus CAMS atmospheric physics across Pakistan.

Monitoring Node	Champion ML Model	Year R ²	Year PM2.5	Year MAE	Year MAPE	Operational Decision Role
Islamabad Campus	Gradient Boosting	0.7973	20.37 ug/m3	12.48 ug/m3	34.63%	Production Forecaster (+1h to +24h)
Karachi Campus	XGBoost	0.7729	14.89 ug/m3	7.89 ug/m3	22.96%	High-Speed Marine Forecaster
Lahore Station	Gradient Boosting	0.8338	42.40 ug/m3	24.15 ug/m3	41.85%	Severe Inversion Training Ground
Hardware Screener	Isolation Forest	N/A	Outliers: 7.6%	Contam: 0.05	Filter Layer	Pre-inference Sensor Fault Detection

4. Core Institutional & Business Benefits to BIC

Strategic BIC Pillar	Direct Operational & Institutional Value to Beaconhouse
Duty of Care & Student Health	Protects thousands of students and faculty with 24-hour predictive smog alerts. Proactively prevents asthma attacks and respiratory illness, reducing absenteeism.
HVAC Energy & Filter Savings	Enables automated modulation of building air dampers during severe pollution hours, extending expensive air filter life and cutting HVAC energy waste by up to 18%.
Brand Leadership & Admissions PR	Positions BIC as <i>'Pakistan's First AI-Powered Smart and Healthy Campus'</i> , creating a compelling, premium differentiator for student enrollment and open-house events.
Academic Distinction & Partners	Demonstrates applied research excellence to UK degree-awarding university partners, proving BIC students deploy live, production-grade AI systems.
Full Asset & IP Ownership	BIC retains 100% ownership of the rooftop hardware stations, proprietary environmental datasets, and trained machine learning models.

5. Verified Itemized Pilot Budget (Updated August 2026)

Component Group / Cost Statement	Items Included & Specifications	Primary Sourcing Channel	Cost (PKR)
Core Sensing & Control Build (Items 1 to 10, 12 to 13)	PMS7003 Laser Sensor, ESP32 WROOM-32D, BME280, MicroSD Module, Samsung 16GB, IP65 Box, 120pc Jumper Kit, 5V 2A PSU, Cable Ties, Hose Clamps, Dual Rain Sensors A & B	Embedded Studio, Digilog.pk, Electrobes, A.E Solution, Daraz	9,660
Validation & Rooftop Additions (Items 14 to 17)	UNI-T UT363 Anemometer, Clopal 10m Heavy-Duty 5-Way Extension Lead, GMSA RTV Silicone (310ml), M3-M6 Stainless Fasteners (8 pcs)	Electrobes, Clopal Online, Expert Tools, Multan Electronics	8,127
Baseline Hardware Subtotal	16 Verified Line Items (Excludes radiation shield pending local quote)	100% Pakistan Sourced	17,787
Contingency Reserve (~5%)	Rounded buffer covering minor price movements at time of order	Budget Reserve	889
TOTAL BASELINE BUDGET REQUEST	Recommended Approval Amount (Single Campus Deployment)	Verified Vendors	PKR 18,676
PROJECT APPROVAL TOTAL	Accommodates Canadian Extension Part 1 Local Radiation Shield Quote	Zero Reporting Fees	PKR 20,000

6. Deliverables at Pilot Conclusion

At the conclusion of the 3-month deployment, the joint team will deliver: (1) Verified 3-month continuous local air quality dataset owned by BIC; (2) Calibrated multi-horizon forecasting models (+1h, +3h, +6h, +24h); (3) Operational HVAC and campus activity decision playbook; (4) Pilot completion executive report with nationwide Beaconhouse expansion roadmap.

7. Formal Management Authorization and Sign-Off

Head of Institute: _____ Ms. Saba Ahson Beaconhouse International College, Islamabad	Head of CSSE / AI (Islamabad): _____ Ms. Sahifa Alam Beaconhouse International College, Islamabad
Project Lead (Islamabad): _____ Munim Qureshi AirSense Project Lead	Project Lead (Karachi): _____ Areesha Aqeel AirSense Project Lead